Data Communication

Content 02: Network Models

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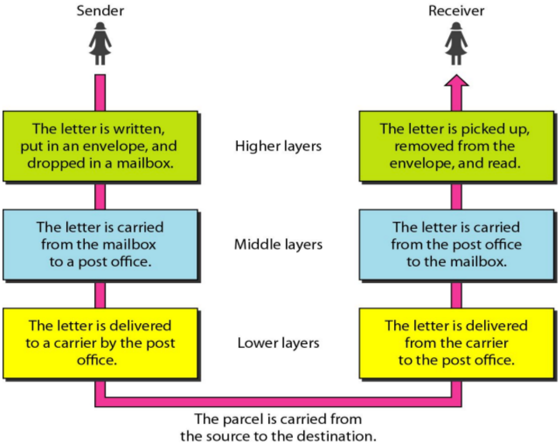
Layered Tasks

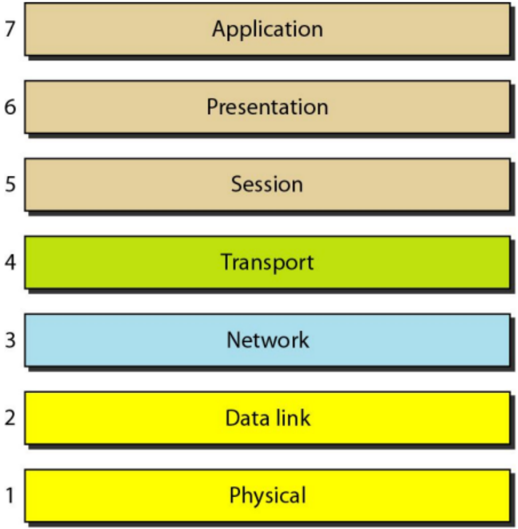
Layered architecture is used to simplify design, enhance interoperability, and allows modular troubleshooting and development.



OSI Model

- **ISO (International Standards Organization)** sets global standards, including network communication standards.
- **OSI (Open Systems Interconnection) model**, introduced in the 1970s, standardizes network communication.
- The OSI model consists of **seven layers**: Physical (Layer 1), Data Link (Layer 2), Network (Layer 3), Transport (Layer 4), Session (Layer 5), Presentation (Layer 6), and Application (Layer 7).
- Designed by grouping related functions into **discrete layers** for standardization.
- Ensures **interoperability** between different network systems.
- Each layer **relies on services** from the layer below it.

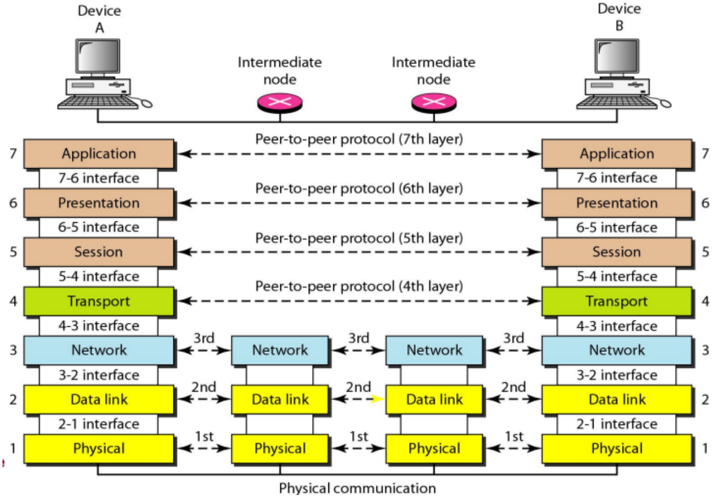
OSI Layered Architecture



Peer-to-Peer Processes

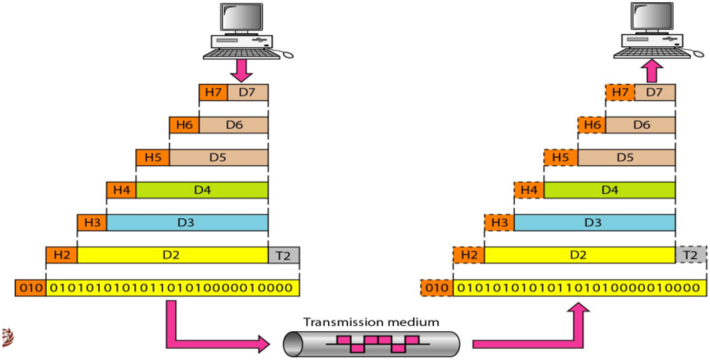
- **Peer-to-Peer Processes**: Layer x on one machine communicates with layer x on another machine.
- **Interfaces Between Layers**: Each interface defines the information and services a layer provides to the layer above it.
- **Organization of Layers**:
  - **Network Support Layers** (Layers 1, 2, 3) handle data transmission and connectivity.
  - **User Support Layers** (Layers 5, 6, 7) manage user interaction and data representation.
  - **Transport Layer** (Layer 4) acts as a **link** between the two subgroups.
- Well-defined interfaces and layer functions ensure **modularity** in a network and enables **interoperability** between unrelated software systems.

Peer-to-Peer Processes



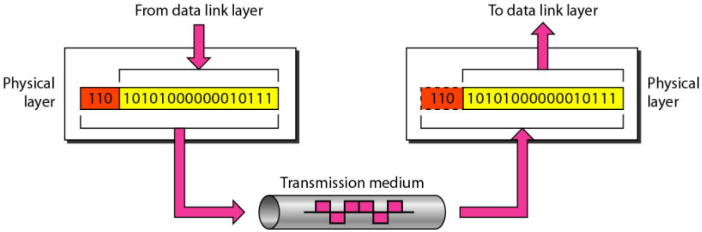
Peer-to-Peer Processes

Encapsulation is the process of wrapping data with protocol-specific headers (and sometimes trailers) as it moves through the layers of a network model.



Layers: Physical Layer

- Coordinates functions required to transmit a bit stream over a physical medium.
- Responsible for movements of individual bits from one hop (node) to the next.



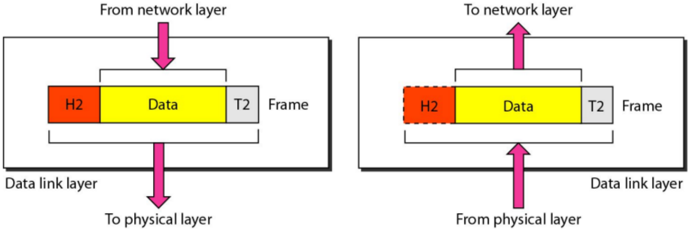
Layers: Physical Layer

Physical layer is concerned and dealt with the mechanical and electrical specification of the primary connections: cable, connector.

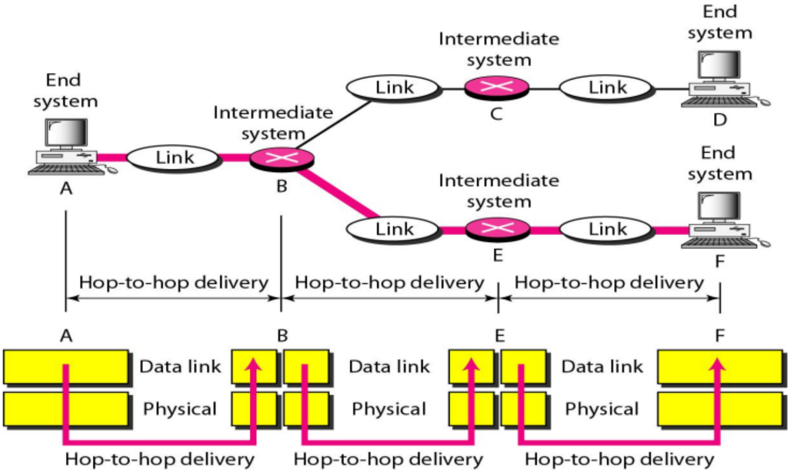
- Physical characteristics of interfaces and medium
- Representation of bits
- Data rate : transmission rate
- Synchronization of bits
- Line configuration
- Physical topology
- Transmission mode

Layers: Data Link Layer

The data link layer is responsible for moving frames from one hop (node) to the next.



Layers: Data Link Layer



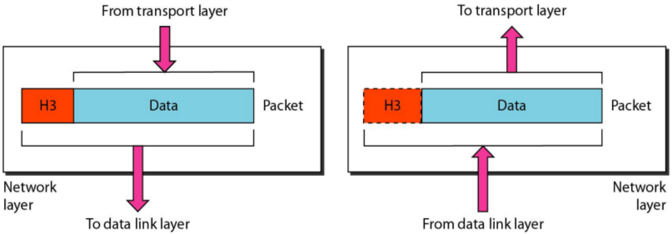
Layers: Data Link Layer

Responsibilities of Data Link Layer:

- Hop-to-hop (node-to-node) delivery
- Framing
- Physical addressing
- Flow control
- Error control
- Access control

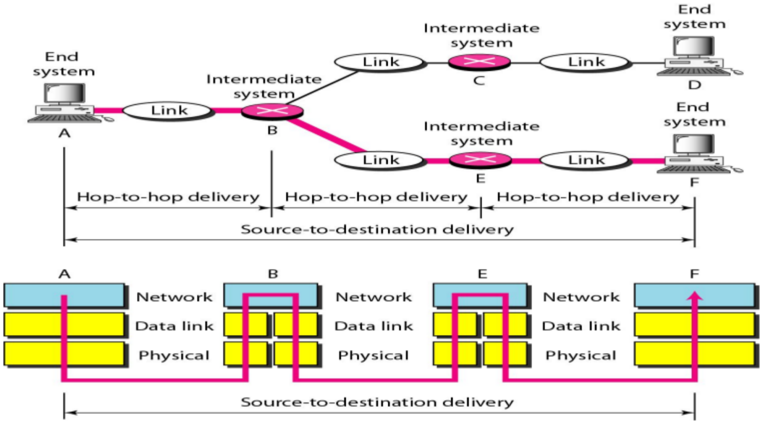
Layers: Network Layer

The network layer is responsible for the delivery of individual packets from the source host to the destination host.



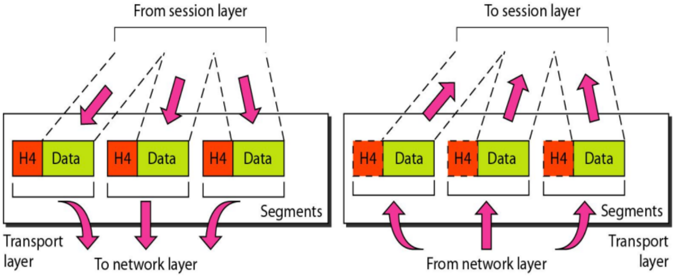
Layers: Network Layer

Responsibilities: **Logical Addressing** and **Routing**.

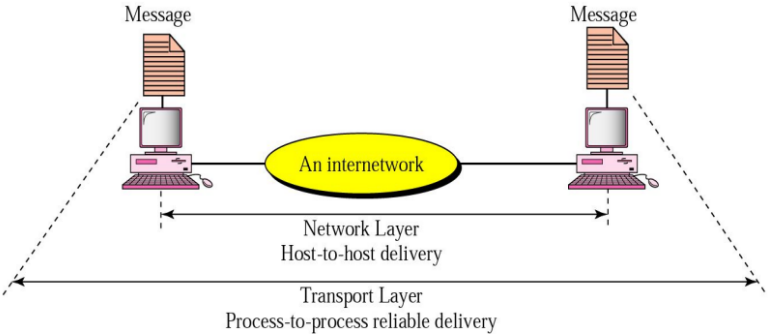


Layers: Transport Layer

The transport layer is responsible for the delivery of a message from one process to another.



Layers: Transport Layer



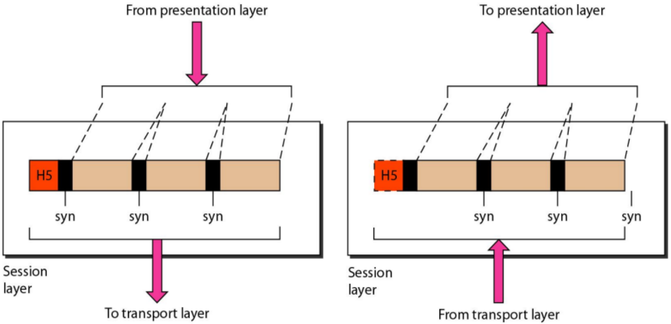
Layers: Transport Layer

Responsibilities of Transport Layer:

- Service port addressing
- Segmentation and reassembly
- Connection control
- Flow control
- Error control

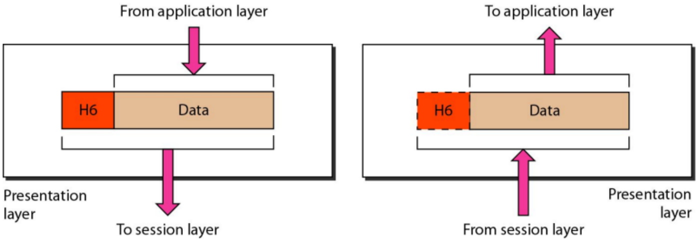
Layers: Session Layer

The session layer is responsible for dialog control and synchronization.



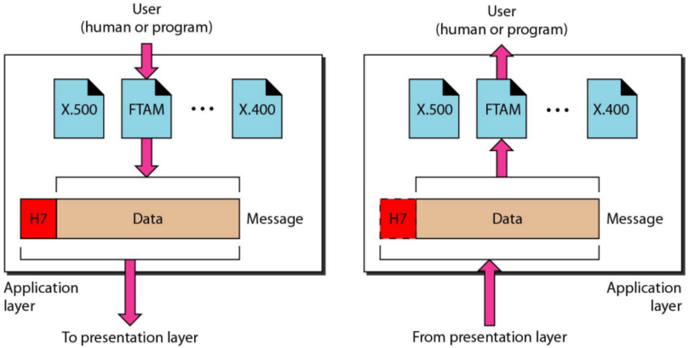
Layers: Presentation Layer

The presentation layer is responsible for translation, compression, and encryption.



Layers: Application Layer

The application layer is responsible for providing services to the user.

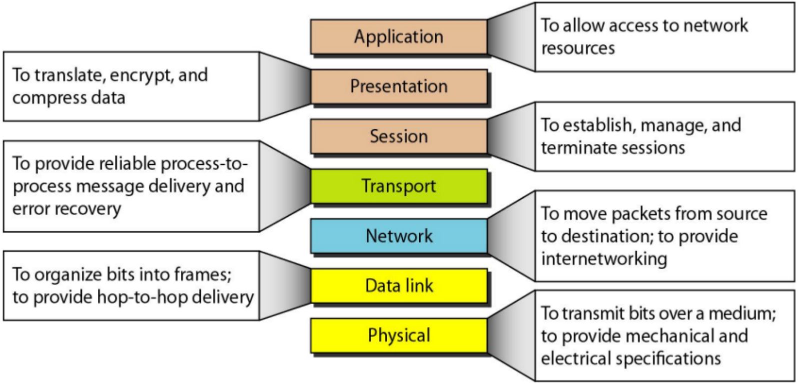


Layers: Application Layer

Responsibilities of the application layer:

- End-User Services
- Resource Sharing & Access
- Network Process to Application

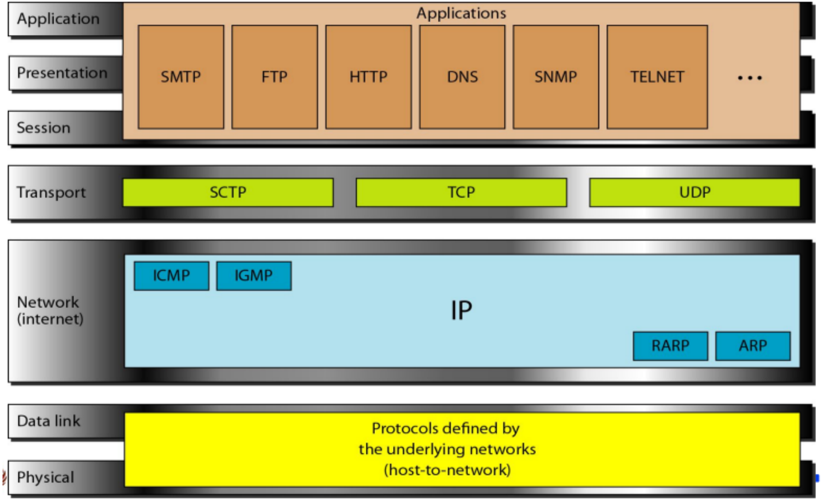
Summary of Layers



TCP/IP Protocol Suite

- **Simplified Model**: A five-layer architecture designed for the internet.
- **Reliable & Unreliable Transport**: Uses TCP (connection-oriented) and UDP (connectionless).
- **IP for Addressing & Routing**: Ensures packet delivery across networks.
- **Core Protocols**: Includes HTTP, FTP, SMTP, DNS, and DHCP for application services.
- **Scalability & Interoperability**: Supports both IPv4 and IPv6.

TCP/IP Protocol Suite

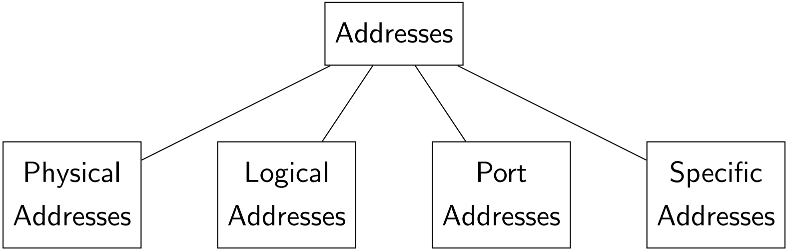


OSI vs TCP/IP

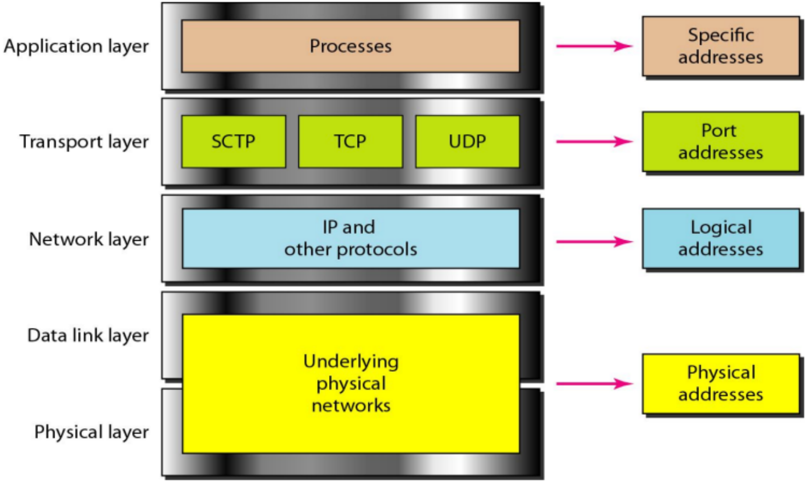
Aspect	OSI Model	TCP/IP Model
Layers	7 layers	5 layers
Structure	Well-defined, modular	Simpler, more flexible
Application Support	Has separate Presentation and Session layers	Merges Application, Presentation, and Session into one layer
Standardization	Conceptual model, not implementation-specific	Developed for real-world communication
Usage	Theoretical, used for teaching	Practical, used in the internet
Addressing	Uses both MAC and IP addresses explicitly	Primarily relies on IP addressing
Protocol Dependency	Protocol-independent model	Tightly bound to TCP and IP protocols

Addressing

Addressing refers to the system of assigning unique identifiers to devices and services to ensure accurate data routing and communication across networks.



Addressing



Application layer
Processes
Specific addresses
Transport layer
SCTP
TCP
UDP
Port addresses
Network layer
IP and other protocols
Logical addresses
Data link layer
Physical layer
Underlying physical networks
Physical addresses

Addressing

The physical addresses will change from hop to hop, but the logical and port addresses usually remain the same.

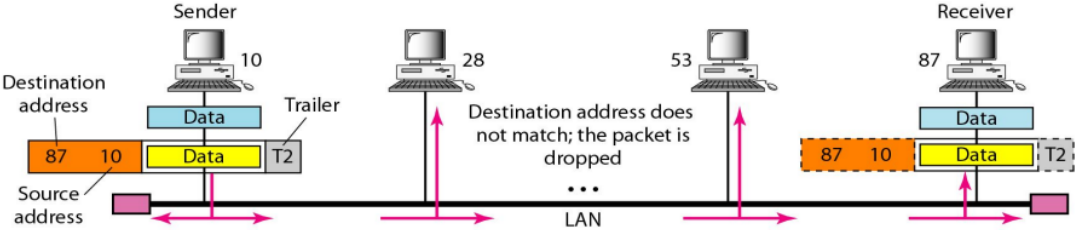
Physical Addresses

- **Physical addresses** are unique identifiers assigned to network interfaces for communication within a local network.
- They are typically **hardware addresses** (e.g., **MAC addresses**) that operate at the **Data Link Layer** of the TCP/IP model.
- These addresses are used for **direct communication** between devices on the same network segment.

Physical Addresses

- A **MAC (Media Access Control) address** is a unique identifier assigned to a network interface card (NIC) for communication within a local network.
- It operates at the **Data Link Layer** of the OSI model and is used for **local addressing and data delivery** within the same network.
- A MAC address is typically represented as a 48-bit hexadecimal number, such as **00:14:22:01:23:45**.
- Example: A device's MAC address allows a switch to forward data to the correct device in a LAN.

Physical Addresses



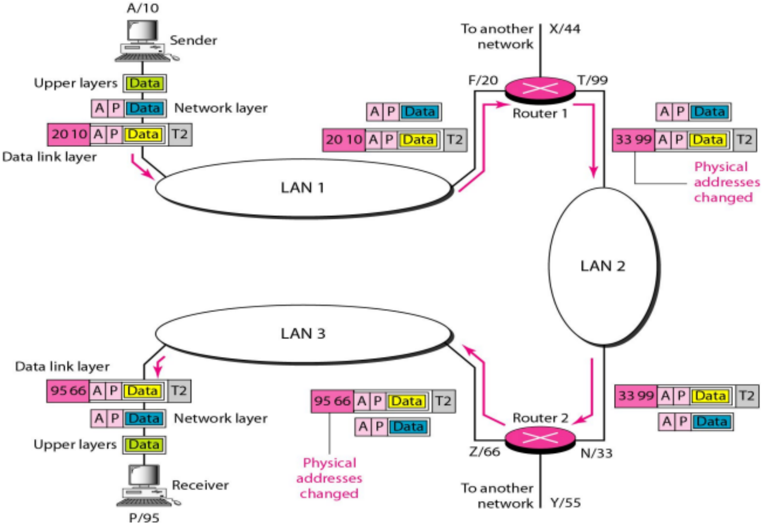
Logical Addresses

- **Logical addressing** refers to the assignment of unique identifiers to devices across different networks.
- It is typically implemented using **IP addresses** at the **Internet Layer** in the TCP/IP model.
- Logical addresses are used for **routing data** between different networks and devices.
- Example: **192.168.1.1** is a typical IPv4 address used to identify a device within a network.

Logical Addresses

- An **IP address** is a unique identifier assigned to devices on a network for **logical addressing** and routing.
- IP addresses are used to identify devices globally and can be either **IPv4 (32-bit)** or **IPv6 (128-bit)**.
- Example: An IPv4 address such as **192.168.1.1** or an IPv6 address: **2001:0db8:85a3:0000:0000:8a2e:0370:7334** is used to uniquely identify a device within a network.

Logical Addresses



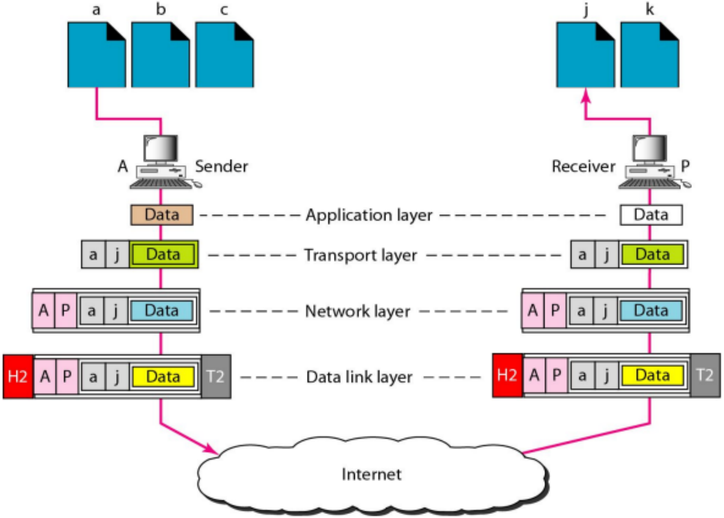
Port Addresses

- A **port address** is used to identify specific applications or services on a device.
- Port addresses work at the **Transport Layer** in the TCP/IP model.
- They are represented by a **16-bit number**, ranging from **0 to 65535**.
- Example: **Port 80** is used for HTTP (web browsing), and **Port 25** is used for SMTP (email).

Port Addresses

- A **port address** identifies specific services or applications on a device by associating a 16-bit number with each communication.
- Port addresses are used in conjunction with IP addresses to ensure that data reaches the correct service on a device.
- Example: **Port 443** is used for secure HTTPS communication.

Port Addresses



Specific Addresses

- A **specific address** refers to a unique identifier for a particular resource, such as an email address or URL.
- Example: An **email address** like **user@example.com** or a **URL** like **http://www.example.com**.

Practice Problems

- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
  Chapter 02 - 4, 5, 9, 10, 11, 12, 17, 20, 21

References

- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
  Chapter 02 - 2.1, 2.2, 2.3, 2.4, 2.5

Thank You